Maha Vishnu

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PROFESSIONAL SUMMARY

Software Development Engineer (SDE) | Full-Stack | React, Node.js, Python, AWS
Cut API latency by 40%, raised throughput by 30%, and reduced production defects by 25% as an SDE intern, while conducting 15+ code reviews per sprint. Builds and ships end-to-end products: a real-time marketplace with payments, a GenAI assistant, and a blockchain verification system with sub-3-second checks. 2026 B.Tech CSE graduate (Amrita Vishwa Vidyapeetham) seeking an entry-level SDE role to build secure, user-focused software.

EDUCATION

B.Tech – Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Coimbatore, India — Sep 2022 – Jun 2026

EXPERIENCE

Software Engineer (R&D) | Garuda Aerospace — ***Apr 2026 – Jul 2026***

- Added GenAI to QGroundControl (QGC), letting operators run missions and query flight data in plain English, cutting menu navigation time by ~40%.
- Built an agentic AI workflow that plans tasks, calls tools, and verifies results, reducing manual ground-control steps by ~30%.
- Secured AI commands with structured JSON outputs, input validation, and fallbacks, reaching ~92% valid-command accuracy across 150 test prompts.

Student Developer | Amrita Student Association — ***Sep 2025 – Jan 2026***

- Built CampusCafe, a university marketplace web app, using React, Vite, Node.js, Express, and MongoDB.
- Added JWT + OTP login, role-based access, and real-time chat and order updates with WebSockets.
- Integrated Razorpay payments and an in-app wallet, and built admin analytics with MongoDB aggregation.
- Created a GenAI food assistant using Flask and a local Ollama LLM.

SDE Intern | Holistic MD LLP — ***Jun 2025 – Sep 2025***

- Increased API throughput by 30%+ by building marketplace modules (React, Node.js, Express).
- Cut API latency by 40% and production defects by 25% by applying secure coding practices (JWT, RBAC, input validation) and SQL query tuning.
- Raised code quality by conducting 15+ code reviews per sprint (Git).
- Wrote unit tests and debugged issues across staging and production with the DevOps team, supporting smooth releases (CI/CD, Git).

PROJECTS EXPERIENCE

CampusCafe

- Designed a 3-tier web app with a React frontend, Express REST API, and MongoDB database, using MVC structure for clean, maintainable code.
- Combined real-time updates, online payments, admin analytics, and a GenAI food assistant in one platform (WebSockets, Razorpay, Flask, Ollama).

Virtual Classroom

- Built a Classes 6–12 learning platform with reusable React components and multi-page routing, backed by an Express API and MongoDB (React, Express, Mongoose).
- Developed 5 learning games with randomized scoring and instant feedback, and set up groundwork for JWT login and file uploads (React, JWT, Multer).

Counterfeit Product Verification System

- Built a product verification system that checks authenticity using image hashing, IPFS storage, and Ethereum smart contracts (Python, Flask, Web3.py).
- Connected blockchain, IPFS, and AWS S3 through REST APIs, using caching and presigned URLs to complete end-to-end verification in under 3 seconds.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java
Frameworks: React.js, Node.js, Django
Databases: MongoDB, MySQL, SQLite
Tools: AWS (S3), Git, GitHub, Postman, Linux, Jira, Confluence, QGroundControl, MAVLink
Concepts: Data Structures & Algorithms, OOP, REST APIs, Secure Coding, Unit Testing, Debugging, SDLC, Agile/Scrum, CI/CD, Generative AI (LLM integration), Agentic AI, Prompt Engineering